FIT5230 Malicious AI

Adversarial Machine Learning I

Overview

- Benign vs Adversarial: attacks on INTegrity
- Semantic adversarial attack
- Noise attack
- Fast Gradient Sign (FGS)
- Fast Gradient Value (FGV)
- Zeroth-Order Optimization (ZOO)
- Recent adversarial attacks on AI

Security attacks AI

- AI with benign samples
  - all samples correct or (at worst) have random errors

vs

- AI with malicious samples
  - some corrupted samples s.t.
    - bias the learning outcome
    - designed to be undetectable/innocent-looking/indistinguishable.
    - **What's the purpose of this?**

Security attacks AI

- AI without adversarial attack: samples may be changed due to errors
  - learning outcome will change in a random manner
  - e.g. class 1 samples misclassified as class 2, class 2 samples misclassified as class 1

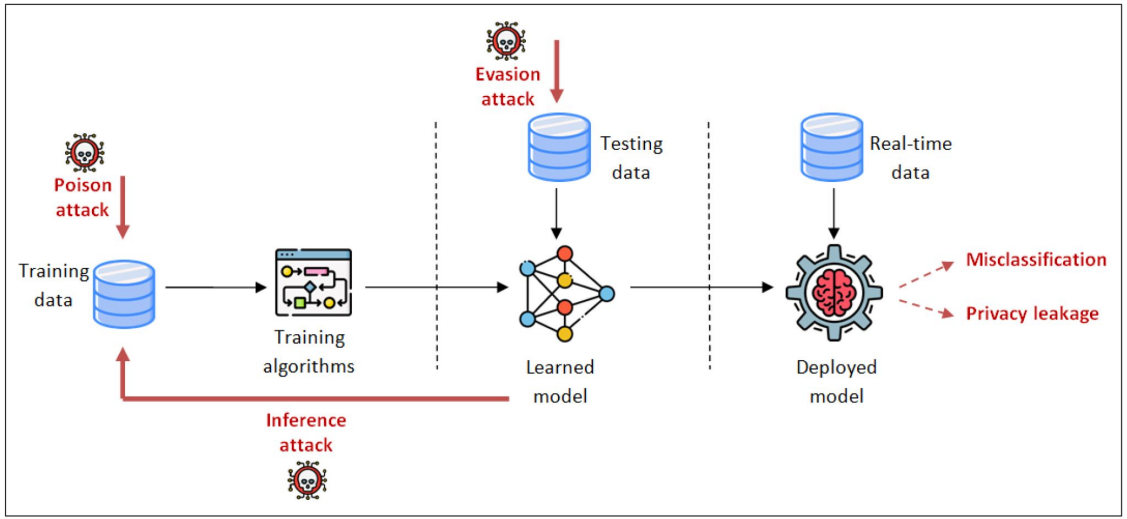
- AI with adversarial attack: samples intentionally corrupted
  - learning outcome will be biased / targeted

Adversarial Attack Classifications

- By attacker knowledge
  - White-box: Full model access, e.g., FSGM
  - Black-box: Only API queries, e.g., Zeroth-Order Optimization
- By goal
  - Targeted: Force classification to a **specific** class, e.g., cat → dog
  - Untargeted: Cause **any** misclassification, e.g., cat → not cat

Adversarial Attack Classifications

- By timing
  - Poisoning: Corrupt training data, e.g., biased hiring algorithm.
  - Evasion: Fool model at test time, e.g., adversarial stop signs.



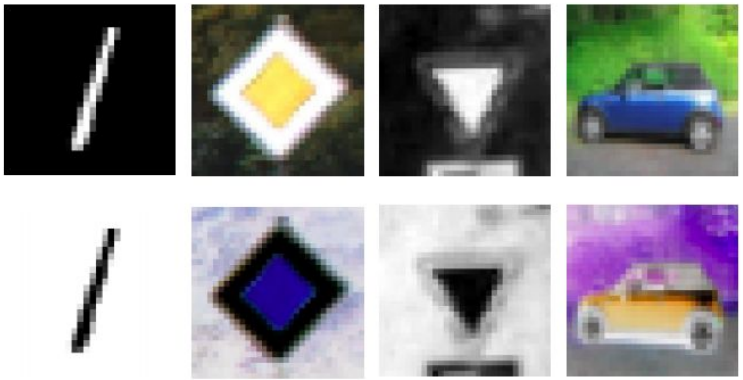
https://www.mdpi.com/1424-8220/24/16/5150

Semantic Adversarial Attack

On the Limitation of Convolutional Neural Networks in Recognizing Negative Images

Hossein Hosseini, Baicen Xiao, Mayoore Jaiswal and Radha Poovendran
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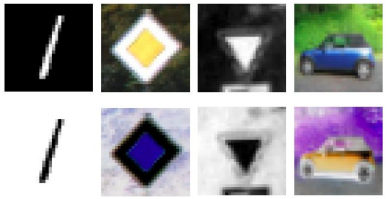
***Abstract*—Convolutional Neural Networks (CNNs) have chieved state-of-the-art performance on a variety of computer ision tasks, particularly visual classification problems, where ew algorithms reported to achieve or even surpass the human erformance. In this paper, we examine whether CNNs are apable of learning the semantics of training data. To this end, e evaluate CNNs on negative images, since they share the same tructure and semantics as regular images and humans can lassify them correctly. Our experimental results indicate that hen training on regular images and testing on negative images, ie model accuracy is significantly lower than when it is tested n regular images. This leads us to the conjecture that current raining methods do not effectively train models to generalize the**



https://arxiv.org/pdf/1703.06857.pdf

Semantic Adversarial Attack

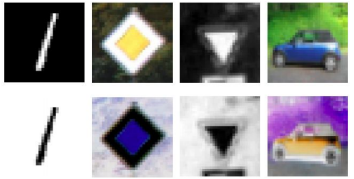
- Can Convolutional Neural Network (CNN) learn semantics of training samples?
- Gist of Semantic adversarial samples: semantically the same as original, but otherwise quite different, e.g. negative images



- Negative image:
  - share same semantics & structure as regular image, humans can easily classify
  - reversed brightness, large pixel-wise perturbation

Semantic Adversarial Attack

- Negative image:
  - reversed brightness, large pixel-wise perturbation, though semantically same
  - e.g. consider a grayscale pixel:
    - 0 = black, (max) 255 = white
    - reverse the brightness: max-pixelvalue
      - black→white: 0 → 255-0 = 255
      - white→black: 255 → 255-255 = 0



- **Q**: is the semantic adversarial attack
  - an untargeted or targeted attack?
  - a white-box or black box attack?

Fast Gradient Signed (FGS) Method

- Goodfellow et al. Explaining & Harnessing Adversarial Examples @ICLR 2015
- Jul 2024: 21192 citations
- Jul 2025: 26070 citations



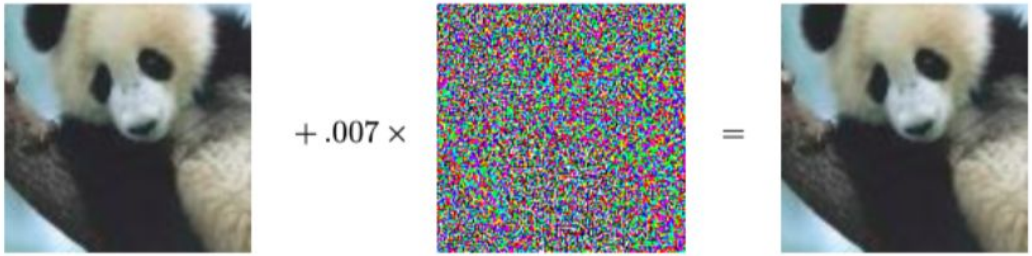
Ian Goodfellow

DeepMind
Verified email at deepmind.com
Deep Learning

Explaining and **harnessing adversarial** examples
IJ Goodfellow, J Shlens, C Szegedy - arXiv preprint arXiv:1412.6572, 2014 - arxiv.org
... of generating **adversarial** examples that makes **adversarial** training practical. We show that **adversarial** ... We start with **explaining** the existence of **adversarial** examples for linear models. ...
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Fast Gradient Sign (FGS) Method

- a white box attack:
  - attacker has complete access to the victim model
- add small perturbations/distortions δ until the classifier labels it as different class/category:
  - $x' = x + \delta$



- **Q**: Is FGS method a poisoning or evasion attack?

Fast Gradient Sign (FGS) Method

- perturbation: use gradient ∇ of the loss J wrt the input image $\boldsymbol{x}$, aiming to maximize that loss J
  - **Q**: why maximize?

- $\boldsymbol{x}' = \boldsymbol{x} + \delta = \boldsymbol{x} + \varepsilon$*sign($\nabla_{\boldsymbol{x}}$J(θ,$\boldsymbol{x}$,y))
  - **Q**: Why do we calculate the loss value if only the sign matters?

  - y = original label of input image $\boldsymbol{x}$
  - ε = multiplier (learning rate) to keep the perturbation small
  - θ = model parameters
  - J = loss

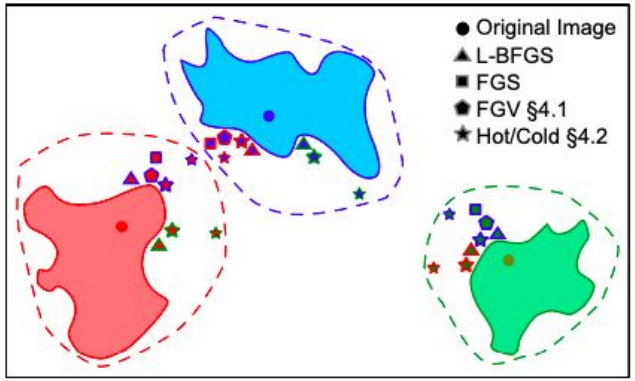
Fast Gradient Value (FGV) Method

Adversarial Diversity and Hard Positive Generation

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Abstract

State-of-the-art deep neural networks suffer from a fundamental problem – they misclassify adversarial examples formed by applying small perturbations to inputs. In this paper, we present a new psychometric perceptual adversarial similarity score (PASS) measure for quantifying adversarial images, introduce the notion of hard positive generation, and use a diverse set of adversarial perturbations – not just the closest ones – for data augmentation. We introduce a novel hot/cold approach for adversarial example generation, which provides multiple possible adversarial pertur-



https://arxiv.org/abs/1605.01775

Fast Gradient Value (FGV) Method

- perturbation: use gradient ∇ of the loss J wrt the input image x, aiming to maximise that loss

- $x' = x + \delta = x + \varepsilon * \text{sign}(\nabla_x J(\theta, x, y))$ $\leftrightarrow$ FGS

Each pixel change the same magnitude (either $+\varepsilon$ or $-\varepsilon$)

- $x' = x + \delta = x + \varepsilon * \nabla_x J(\theta, x, y)$ $\leftrightarrow$ FGV

Scales the perturbation proportionally to the pixel's contribution to the loss

Zeroth-Order Optimization (ZOO) Method

ZOO: Zeroth Order Optimization Based Black-box Attacks to Deep Neural Networks without Training Substitute Models

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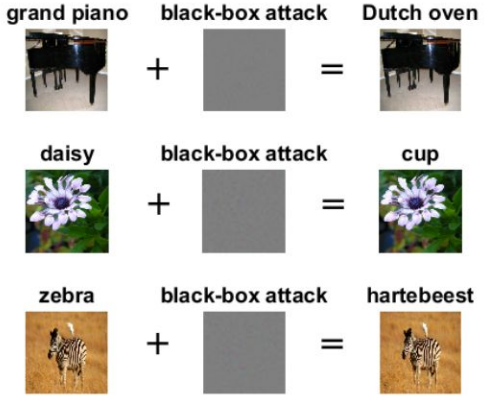
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ABSTRACT

Deep neural networks (DNNs) are one of the most prominent technologies of our time, as they achieve state-of-the-art performance in many machine learning tasks, including but not limited to image classification, text mining, and speech processing. However, recent research on DNNs has indicated ever-increasing concern on the robustness to adversarial examples, especially for security-critical

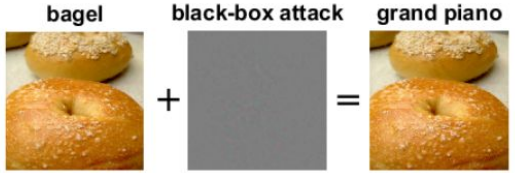


Zeroth-Order Optimization (ZOO) Method

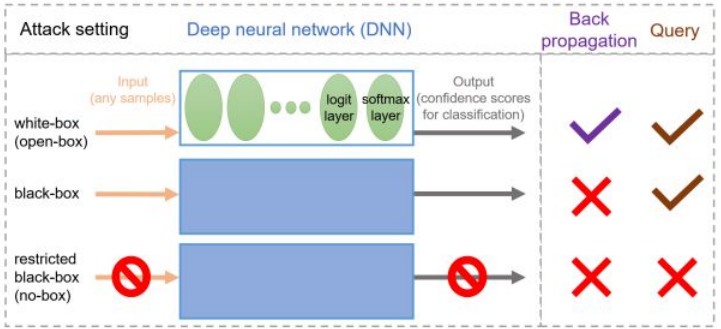
- a black-box attack
- perturbation: change pixel values of the input.
- query model: ask black-box for outputs
  - Use output differences to approximate gradients:

$$\hat{\nabla}_i f(x) \approx \frac{f(x + \delta e_i) - f(x - \delta e_i)}{2\delta}$$

- Adjust input to maximize error to cause misclassification.



(a) a ZOO black-box targeted attack example



Activity 1

Classify each scenario by:

- Attacker knowledge (White-box or Black-box)
- Goal (Targeted or Untargeted)
- Timing (Poisoning or Evasion)

Scenarios:

1. A self-driving car misreads a stop sign due to stickers.
2. Training data is poisoned to bias hiring outcomes.
3. An API-based translation service is probed to create adversarial sentences.
4. Slight pixel changes cause an image classifier to mislabel cats as dogs.
5. Spam emails are designed to bypass filters by mimicking legitimate structures.

Provide your answer here: https://shorturl.at/8PrwW

Activity 2

The Thirty-Seventh AAAI Conference on Artificial Intelligence (AAAI-23)

CodeAttack: Code-Based Adversarial Attacks for Pre-trained Programming Language Models

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Abstract

Pre-trained programming language (PL) models (such as CodeT5, CodeBERT, GraphCodeBERT, etc.,) have the potential to automate software engineering tasks involving code understanding and code generation. However, these models operate in the natural channel of code, *i.e.*, they are primarily concerned with the human understanding of the code. They are not robust to changes in the input and thus, are potentially susceptible to adversarial attacks in the natural channel. We propose, **CodeAttack**, a simple yet effective black-box attack model that uses code structure to generate effective, efficient, and imperceptible adversarial code samples and demonstrates the vulnerabilities of the state-of-the-art PL models to code-specific adversarial attacks. We evaluate the transferability of CodeAttack on several code-code (translation and repair) and code-NL (summarization) tasks across different programming languages. CodeAttack outperforms state-of-the-art adversarial NLP attack models to achieve the best overall drop in performance while being more efficient, imperceptible, consistent, and fluent. The code can be found at https://github.com/reddy-lab-code-research/CodeAttack.

https://doi.org/10.1609/aaai.v37i12.26739

Read the paper and answer the following questions:

1. What is the core problem addressed in this paper?
2. What is the main contribution or solution proposed by the authors?
3. What is the primary conclusion drawn from the paper's findings?
4. Are the code and resources, such as a GitHub repository, provided for the proposed method?
5. Does the paper's methodology for generating adversarial attacks include sufficient detail for one to replicate the results?
6. Are there clear instructions on how to set up the environment and run the experiments?
7. Does the paper specify the datasets used for evaluation, and are these datasets publicly available?
8. Does the paper provide a thorough and convincing evaluation of the proposed method's performance?

Provide your answer here: https://shorturl.at/gZJHF